Strategies for Enhancement in Food Production: A Comprehensive Study Guide

As the global population continues to rise, the enhancement of food production has become a critical necessity. This objective is met through the application of biological principles to animal husbandry and plant breeding, alongside modern techniques such as embryo transfer and tissue culture.

🌾 CHAPTER 9: STRATEGIES FOR ENHANCEMENT IN FOOD PRODUCTION

With the ever-increasing population of the world, enhancement of food production is a major necessity. By applying biological principles to animal husbandry and plant breeding, and through new techniques like tissue culture and embryo transfer, we can vastly increase our food yield.

🐄 TOPIC 1: ANIMAL HUSBANDRY

Animal husbandry is the agricultural practice of breeding and raising livestock (such as cows, buffaloes, pigs, horses, sheep, and poultry) that are useful to humans.

1. Management of Farms and Farm Animals

- **Dairy Farm Management (Dairying):** The management of animals for milk and its products for human consumption.
    - **Key factors for high yield:** Selection of good breeds with high yielding potential and resistance to diseases.
    - **Care:** Cattle must be housed well, given adequate water, maintained disease-free, and fed scientifically (quality and quantity of fodder). Stringent cleanliness and hygiene are of paramount importance.
- **Poultry Farm Management:** Rearing domesticated fowls (chicken, ducks, turkeys, geese) for their meat and eggs. It requires disease-free breeds, proper feed/water, and safe farm conditions.

2. Animal Breeding

The goal of animal breeding is to increase the yield of animals and improve the desirable qualities of their produce.

- **Breed:** A group of animals related by descent and similar in most characters (general appearance, features, size).

Flow Chart: Types of Animal Breeding

- ANIMAL BREEDING
-     │
-     ├──> A. INBREEDING (Mating within the same breed)
-     │
-     └──> B. OUTBREEDING (Mating between unrelated animals)
-             │
-             ├──> 1. Out-crossing (Same breed, no common ancestors)
-             │
-             ├──> 2. Cross-breeding (Different breeds)
-             │
-             └──> 3. Interspecific Hybridisation (Different species)

A. Inbreeding

- **Definition:** Mating of more closely related individuals within the **same breed for 4-6 generations**.
- **Advantages:** It increases **homozygosity**, which is necessary to evolve a pureline. It exposes harmful recessive genes so they can be eliminated by selection and helps accumulate superior genes.
- **Disadvantage (Inbreeding Depression):** Continued close inbreeding reduces fertility and productivity.
- *Solution:* Mate the depressed animals with unrelated superior animals of the same breed (a single outcross) to restore fertility and yield.

B. Outbreeding

- **1. Out-crossing:** Mating of animals within the same breed, but having **no common ancestors on either side of their pedigree up to 4-6 generations**. It is the best breeding method for animals with below-average productivity in milk or beef.
- **2. Cross-breeding: Superior males of one breed are mated with superior females of another breed**.
    - *Example:* **Hisardale** is a new breed of sheep developed in Punjab by crossing Bikaneri ewes and Marino rams.
- **3. Interspecific Hybridisation:** Male and female animals of **two different related species** are mated.
    - *Example:* The **mule** (produced by crossing a male donkey with a female horse).

3. Controlled Breeding Experiments

- **Artificial Insemination:** Semen is collected from an elite male parent and injected into the reproductive tract of the selected female. The semen can be used immediately or frozen for later transport. It helps overcome several problems associated with normal mating.
- **MOET (Multiple Ovulation Embryo Transfer Technology):** A programme for herd improvement.
    1. A cow is administered hormones with FSH-like activity.
    2. This induces **superovulation** (producing **6-8 eggs** per cycle instead of the normal one).
    3. The cow is mated with an elite bull or artificially inseminated.
    4. Fertilised eggs at the **8–32 cell stage** are recovered non-surgically and transferred to **surrogate mothers**.

4. Bee-keeping (Apiculture) & Fisheries

- **Apiculture:** Maintenance of hives of honeybees for the production of honey (high nutritive and medicinal value) and beeswax (used in cosmetics and polishes). The most common species is ***Apis indica***. Keeping beehives in crop fields (like apple, Brassica) during the flowering period increases pollination efficiency and crop yield.
- **Fisheries:** Catching, processing, or selling of fish, shellfish, and other aquatic animals.
    - *Freshwater fishes:* Catla, Rohu, common carp.
    - *Marine fishes:* Hilsa, Sardines, Mackerel, Pomfrets.
    - *Note:* The massive increase in aquatic production is termed the **Blue Revolution**.

🌱 TOPIC 2: PLANT BREEDING

Plant breeding is the purposeful manipulation of plant species to create desired plant types that are better suited for cultivation, give better yields, and are disease-resistant.

Flow Chart: Main Steps in Plant Breeding

- 1. Collection of variability (Germplasm collection: gathering all diverse alleles/wild varieties)
- ↓
- 2. Evaluation and selection of parents (identifying plants with desirable traits)
- ↓
- 3. Cross hybridisation among selected parents (combining desired characters from two different plants)
- ↓

- 4. Selection and testing of superior recombinants (selecting progeny superior to both parents and self-pollinating them to homozygosity)
- ↓
- 5. Testing, release, and commercialisation of new cultivars (evaluating yield, quality, and disease resistance in research fields for at least three seasons)

1. The Green Revolution in India

The development of high-yielding varieties of crops in the mid-1960s led to a dramatic increase in food production.

- **Wheat & Rice:** Nobel laureate Norman E. Borlaug developed semi-dwarf wheat. *Sonalika* and *Kalyan Sona* are high-yielding, disease-resistant wheat varieties introduced in India. Semi-dwarf rice varieties *Jaya* and *Ratna* were also developed.
- **Sugarcane:** *Saccharum barberi* (grown in North India, poor sugar yield) was successfully crossed with *Saccharum officinarum* (grown in South India, thick stems, high sugar) to get a hybrid with high yield, thick stems, high sugar, and the ability to grow in North India.
- **Millets:** Hybrid maize, jowar, and bajra successfully developed in India are highly resistant to water stress.

2. Plant Breeding for Disease Resistance

Enhances food production and reduces dependence on fungicides and bactericides.

- **Wheat:** *Himgiri* variety (Resistant to Leaf and stripe rust, hill bunt).
- **Brassica:** *Pusa swarnim* (Resistant to White rust).
- **Cauliflower:** *Pusa Shubhra* & *Pusa Snowball K-1* (Resistant to Black rot).
- **Cowpea:** *Pusa Komal* (Resistant to Bacterial blight).

Mutation Breeding: Inducing mutations artificially using chemicals or radiations (like gamma radiations) to create a new, desirable character not found in parental types, and using these plants in breeding.

- *Example:* In mung bean, resistance to yellow mosaic virus and powdery mildew was induced by mutations.

3. Plant Breeding for Insect Pest Resistance

Morphological, biochemical, or physiological characteristics give plants natural resistance (e.g., hairy leaves in cotton repel jassids).

- **Brassica:** *Pusa Gaurav* (Resistant to Aphids).
- **Okra (Bhindi):** *Pusa Sawani* & *Pusa A-4* (Resistant to Shoot and Fruit borer).

4. Biofortification (Improved Food Quality)

Biofortification is the process of breeding crops with higher levels of vitamins, minerals, proteins, and healthier fats to improve public health and combat "hidden hunger".

- *Examples:*
  - Maize hybrids with twice the amount of amino acids (lysine and tryptophan).
  - *Atlas 66* (a wheat variety with high protein content).
  - Iron-fortified rice (5 times more iron).
  - Vitamin A-enriched carrots, spinach, and pumpkins developed by IARI, New Delhi.

TOPIC 3: SINGLE CELL PROTEIN (SCP)

With the shift from a grain to a meat diet, cereal demand increases drastically (it takes 3-10 kg of grain to produce 1 kg of meat).

- **Single Cell Protein (SCP)** serves as an excellent alternate source of proteins for human and animal nutrition to combat malnutrition.
- **Microbes as Food:** Microbes are grown on an industrial scale. For example, the blue-green alga ***Spirulina*** can be easily grown on wastewater from potato processing plants, straw, molasses, and animal manure.
- **Advantages:** It produces large quantities of food rich in protein, minerals, fats, and vitamins, while simultaneously reducing environmental pollution.
- *Note:* Certain bacteria like *Methylophilus methylotrophus* can produce up to 25 tonnes of protein due to their rapid growth rate.

TOPIC 4: TISSUE CULTURE

When traditional breeding fails to meet demands efficiently, tissue culture is utilized.

- **Totipotency:** The unique capacity to generate a whole plant from any cell or explant (any part of a plant taken out and grown in a test tube) under sterile conditions in a special nutrient media.
- **Micropropagation:** The method of producing thousands of plants in a very short duration using tissue culture.
  - **Somaclones:** Plants produced via micropropagation are genetically identical to the original parent plant, hence they are called somaclones. Commercial plants like tomato, banana, and apple are produced this way.

- **Meristem Culture (Disease-free recovery):** Even if a plant is infected with a virus, its **meristem (apical and axillary) is free of viruses**. Scientists remove the meristem and grow it *in vitro* to successfully recover virus-free, healthy plants (e.g., banana, sugarcane, potato).
- **Somatic Hybridisation:**
    - Scientists isolate single cells from plants and digest their cell walls to get **naked protoplasts**.
    - Isolated protoplasts from two different plant varieties (each with desirable traits) are fused to get **hybrid protoplasts**.
    - These grow into new plants called **somatic hybrids**.
    - *Example:* **Pomato** (Protoplast of potato + Protoplast of tomato). Though successfully created, it unfortunately lacked the exact combination of traits needed for commercial success.

IV. Review Quiz

Instructions: Provide short-answer responses (2-3 sentences) for the following questions.

1. What is the primary difference between inbreeding and out-crossing?
2. How does the MOET program improve the quality of a herd?
3. Why is the selection of "good breeds" considered the most important factor in dairy farm management?
4. Explain the significance of the "Blue Revolution."
5. What are the five main steps involved in breeding a new genetic variety of a crop?
6. Define "biofortification" and provide one specific example of a biofortified crop.
7. How can mutation breeding be used to enhance crop resistance?
8. What is "totipotency" and why is it foundational to tissue culture?
9. Explain how scientists can obtain virus-free plants from a mother plant that is already infected.
10. Describe the potential benefits of producing Single Cell Protein (SCP) using Spirulina.

V. Quiz Answer Key

1. **Inbreeding vs. Out-crossing:** Inbreeding involves mating closely related individuals within the same breed for 4–6 generations to increase homozygosity. Out-crossing is a form of out-breeding where animals of the same breed are mated, but they share no common ancestors for at least 4–6 generations.
2. **MOET Program:** MOET improves herds by inducing super ovulation in a high-quality genetic mother to produce multiple eggs (6–8) per cycle. These embryos are then transferred to surrogate mothers, allowing the genetic mother to be available for another round of breeding quickly.
3. **Selection of Good Breeds:** The quality of the breed determines the milk yield potential and resistance to diseases under specific climatic conditions. Even with perfect management, the maximum yield is limited by the inherent genetic potential of the breed itself.
4. **Blue Revolution:** The Blue Revolution refers to the rapid increase and flourishing of the fishery industry through aquaculture and pisciculture. It is significant for providing income

and employment to millions, especially in coastal states, while meeting the increasing global demand for aquatic food.

5. **Steps in Plant Breeding:** The process begins with the collection of genetic variability (germplasm) and the evaluation/selection of parents. This is followed by cross-hybridisation, the selection and testing of superior recombinants, and finally the testing, release, and commercialisation of new cultivars.
6. **Biofortification:** Biofortification is the process of breeding crops to increase their nutritional value, such as higher vitamin, mineral, or protein content. An example is the development of maize hybrids in 2000 that contained twice the amount of the amino acids lysine and tryptophan.
7. **Mutation Breeding:** Mutation breeding involves inducing genetic changes artificially using chemicals or radiation to create new desirable traits not found in the parental type. These mutants are then screened for traits like disease resistance and used as a source in further breeding programs.
8. **Totipotency:** Totipotency is the inherent capacity of any plant cell or explant to generate a whole new plant when provided with the correct nutrient medium and sterile conditions. It is the core biological principle that allows for micropropagation and the commercial production of somaclones.
9. **Virus-Free Plants:** Even if a plant is systemically infected with a virus, the apical and axillary meristems usually remain virus-free. Scientists can remove these specific tissues and grow them *in vitro* to regenerate entirely healthy, virus-free plants.
10. **Benefits of Spirulina (SCP):** Spirulina provides a high-protein food source rich in minerals and vitamins that can be grown easily on industrial waste like sewage or molasses. This process helps meet the nutritional demands of a growing population while simultaneously reducing environmental pollution.

VI. Essay Questions

Instructions: These questions require in-depth analysis and synthesis of the source material. No answers are provided.

1. Compare and contrast the management practices required for successful dairy farming versus successful bee-keeping.
2. Discuss the economic and social impact of the Green Revolution in India, specifically highlighting the role of semi-dwarf varieties.
3. Evaluate the different methods of animal breeding (inbreeding, out-crossing, cross-breeding, and interspecific hybridisation) and determine which is most effective for a farmer looking to overcome "inbreeding depression."
4. Explain the process and potential challenges of somatic hybridisation, using the example of the "pomato" to illustrate your points.
5. Analyze how plant breeding for disease resistance and insect pest resistance contributes to environmental sustainability in agriculture.

VII. Glossary of Key Terms

Term,Definition

Apiculture,The maintenance of honeybee hives for the production of honey and beeswax; also known as bee-keeping.
Biofortification,"Breeding crops with higher levels of vitamins, minerals, proteins, or healthier fats to improve public health."
Breed,"A group of animals related by descent and similar in most characters like general appearance, features, and size."
Explant,Any part of a plant taken out and grown in a test tube under sterile conditions for tissue culture.
Germplasm Collection,The entire collection of plants or seeds having all the diverse alleles for all genes in a given crop.
Inbreeding Depression,"The reduction in fertility and productivity that results from continued, close inbreeding."
Micropropagation,The method of producing thousands of plants through tissue culture in a very short duration.
Mutation,The process by which genetic variations are created through changes in the base sequence within genes.
Out-breeding,"The breeding of unrelated animals, which may be between different individuals of the same breed, different breeds, or different species."
Single Cell Protein (SCP),An alternate source of protein for human and animal nutrition derived from microorganisms like Spirulina.
Somaclones,Plants produced through tissue culture that are genetically identical to the original plant from which they were grown.
Somatic Hybrids,Hybrid plants formed by the fusion of isolated protoplasts from two different varieties of plants.
Totipotency,The capacity of a single cell or explant to generate a whole new plant under specific laboratory conditions.